

Website design evaluation: reward, metrics, and agent behavior

We built a browser-state evaluator for screenshot-to-code website reproduction. The main finding is clear: broad visual similarity is not enough. The reward needs rendered HTML, CSSOM, visual blocks, and state coverage to distinguish faithful reproduction from a plausible page in the same genre.

12

synthetic websites

Multi-page tasks generated as a first dataset for Harbor evaluation.

153

capture rows

Full-page and stateful captures: dropdowns, filters, tabs, accordions, selected nodes.

0.194-0.678

reward spread

Enough separation to inspect what the reward is actually learning.

Core insight

DreamSim and global pixelmatch often stay high for weak candidates. The metrics that separate good from bad are rendered HTML, CSSOM, visual-block geometry, bbox layout, and state coverage.

1. What we built

The evaluator treats the browser as the source of truth. For each oracle capture, it serves the reference and candidate pages, replays the capture state, then extracts screenshots, rendered DOM/outerHTML, CSSOM, visual blocks, and layout boxes from that live page state.

This matters because future candidates may be plain HTML, React, Solid, Tailwind, or a build system. The reward should not care about source shape. It should care about what the browser renders.

Design principle

The oracle manifest defines what is fair to score. If a dropdown, filter, tab, or selected state is in the oracle manifest, then we evaluate whether the candidate can reproduce that visible state.

Pipeline

1. Generate synthetic oracle website.
2. Capture oracle screenshots and state manifest.
3. Give only screenshots to the coding agent.
4. Agent builds candidate website.
5. Evaluator generates candidate capture plan and replays states.
6. Reward combines coverage, visual, DOM, CSSOM, VLM, and geometry metrics.

The run was analytically valid: metrics completed for all 12 trials. Some trials hit an old markdown report writer bug after metrics completed; that bug does not affect the score data used here.

2. Reward overview

The scalar reward is a weighted score per manifest capture. Coverage is applied once as the outer multiplier. Unsupported evaluator components are removed from the available denominator; numeric zeros remain real zeros.

Component	Weight	Role
Screenshot size match	0.05	Cheap rendering/page-length guard.
Rendered HTML	0.10	Content and DOM structure.
VLM	0.20	Human-like visual judgement.
Pixel match	0.05	Global + blockwise pixel fidelity.
Visual block aggregate	0.00	Reported as evidence; not directly weighted in the current scalar reward.
BBox geometry	0.10	Block shape and placement, derived from visual-block matching.
CSSOM style	0.10	Computed style fidelity over matched visual blocks.
DreamSim	0.10	Broad perceptual similarity.

Gate

A capture must pass screenshot size, rendered HTML, and VLM thresholds before the more specific components contribute. This prevents a page with the wrong content or broken rendering from receiving high credit because one visual metric is high.

3. Trial-level reward results

The spread is meaningful: the best candidate is about 3.5x the lowest reward. DreamSim remains high for weak candidates, while rendered HTML, CSSOM, and visual-block/geometry metrics separate fidelity.

Trial	Reward	Coverage	HTML	CSSOM	VB	Read
site-01-education-k12	0.678	0.957	0.634	0.646	0.855	Best overall. Structure, content, and styling line up.
site-05-retail-marketplace	0.620	0.882	0.507	0.589	0.767	Good visual system; missing density states.
botanical-garden-nursery	0.607	0.975	0.477	0.622	0.831	Strong visual blocks; one state miss.
site-04-science-research	0.599	0.959	0.496	0.452	0.750	Good broad reproduction; weaker style/geometry.
podcast-network	0.596	0.944	0.473	0.509	0.759	Good page vibe; state gaps remain.
storm-chasers-community	0.576	0.882	0.541	0.519	0.748	Decent static structure.
regional-credit-union	0.569	1.000	0.485	0.521	0.809	Coverage complete; local branch-state geometry failed.
tabletop-gaming-cafe	0.556	0.850	0.496	0.494	0.711	Low coverage from missed interaction states.
zen-center	0.548	0.943	0.438	0.575	0.801	High broad visual scores; HTML gate catches drift.
curling-federation	0.414	0.947	0.382	0.503	0.776	Same domain, wrong density/card system.
indie-record-label	0.261	1.000	0.314	0.420	0.768	Plausible vibe, weak rendered content structure.
lutherie-guild	0.194	1.000	0.279	0.271	0.699	DreamSim high, real structure/style low.

Reward honesty check: does 0.6 look better than 0.2?

Reward honesty check: higher reward should look more faithful

High reward example

K-12 school dashboard

Reward 0.678: page grammar, layout density, and rendered structure mostly survive.

High-fidelity institutional dashboard

The user perspective, robust dashboard structure, repeated cards, side nav, and dense content.

Reference / oracle



Full reward: 0.678 - reference: 0.678 - HTML: 0.678 - CSS: 0.678 - JS: 0.678 - Visual: 0.678

Good reproductions keep page grammar, not just color and browser layout.

Candidate



Mid/high reward example

Vintage marketplace browse

Reward 0.620: strong visual system, but density/state details still drift.

Vintage marketplace with commerce density

Commerce look: dense product grids, filters, forms, dense content, consistent conventions.

Reference / oracle



Full reward: 0.620 - reference: 0.620 - HTML: 0.620 - CSS: 0.620 - JS: 0.620 - Visual: 0.620

Agents get the market feel but miss some density-specific details.

Candidate



Low reward example

Lutherie guild home

Reward 0.194: plausible broad mood, weak structure/style/content fidelity.

Craft guild: broad mood is easier than exact grammar

Practiced mood content, but typography hierarchy, copy density, and local structure drift.

Reference / oracle



Full reward: 0.194 - reference: 0.194 - HTML: 0.194 - CSS: 0.194 - JS: 0.194 - Visual: 0.194

This is a useful low reward example: plausible aesthetic, weak HTML reproduction.

Candidate



A quick ordered read: high reward preserves page grammar and local structure; mid/high reward keeps the system but loses some density/state detail; low reward preserves mood while failing concrete reproduction.

Short answer: yes, directionally. The 0.6-0.7 candidates are visibly more faithful than the 0.2 candidates. The high-reward pages preserve the original page type, repeated components, content density, nav/sidebar/footer grammar, and browser-rendered structure. The low-reward pages may still be attractive, but they are closer to a genre rewrite than a reproduction. A 0.6 is not pixel-perfect; it is a candidate that cleared more of the browser-state reproduction checks.

0.6+ retained page grammar

0.4-0.6 broad match with local drift

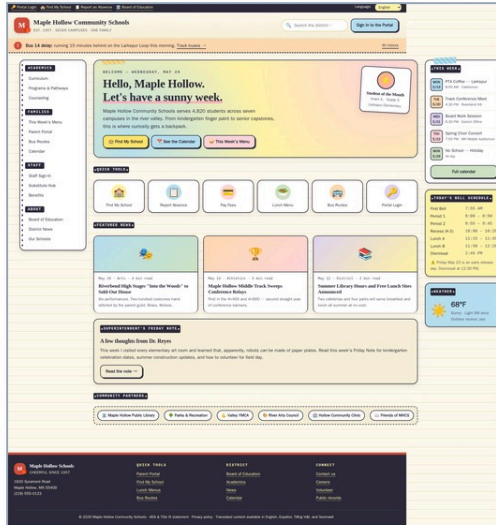
0.2-ish vibe without faithful reproduction

4. Evidence: high reward requires page grammar, not just resemblance

High-fidelity institutional dashboard

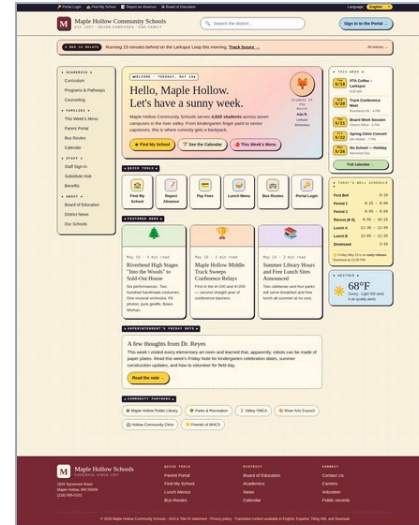
The agent preserves school dashboard structure, repeated cards, side rail, and dense content.

Reference / oracle



trial reward 0.678 · coverage 0.957 · HTML 0.634 · CSSOM 0.646 · visual block 0.855
Good reproductions keep page grammar, not just color and broad layout.

Candidate



K-12 school site: strongest overall candidate. It preserves dashboard-like layout, side rail, dense cards, palette, footer, and enough rendered content structure.

The highest-scoring task is not merely attractive. It keeps the page grammar: repeated structures, content density, navigation, and styling survive the reproduction.

reward 0.678

HTML 0.634

CSSOM 0.646

visual block 0.855

coverage 0.957

5. Evidence: DreamSim over-credits broad aesthetic resemblance

Craft guild: broad mood is easier than exact grammar

Parchment mood survives, but typography hierarchy, copy density, and local structure drift.

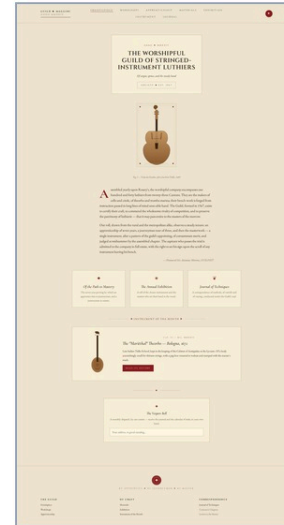
Reference / oracle



trial reward 0.194 · DreamSim 0.849 · HTML 0.279 · CSSOM 0.271 · BBox 0.143

This is a useful low-end example: plausible aesthetic, weak faithful reproduction.

Candidate



Lutherie guild: parchment mood and central composition survive, but actual structure, typography, copy, and layout fidelity are poor.

This is the cleanest warning against making DreamSim a primary reward driver. The broad aesthetic family is right, but the reproduction is not faithful.

reward 0.194

DreamSim 0.849

HTML 0.279

CSSOM 0.271

BBox 0.143

6. Evidence: plausible visual clone, weak rendered structure

Indie record label: visually plausible, structurally weak

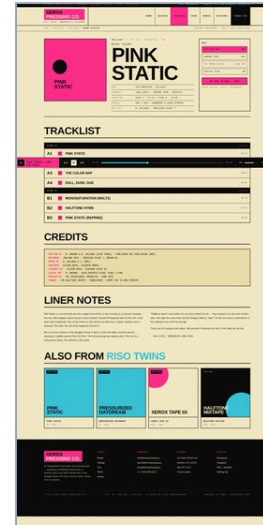
Music/editorial vibe is strong, while rendered content and page-specific structure drift.

Reference / oracle



trial reward 0.261 · VLM 0.652 · visual block 0.768 · HTML 0.314 · CSSOM 0.420
The benchmark separates genre-correct pages from faithful screenshots-to-code.

Candidate



Indie record label: the candidate has the right vibe, but content structure and page-specific details drift.

A judge or perceptual model can be tempted by a plausible page. Rendered HTML and CSSOM prevent over-crediting by checking whether the actual browser-rendered structure is close.

reward 0.261

VLM 0.652

visual block 0.768

HTML 0.314

CSSOM 0.420

7. Evidence: visual block works when visible blocks align

Stateful botanical map reproduction

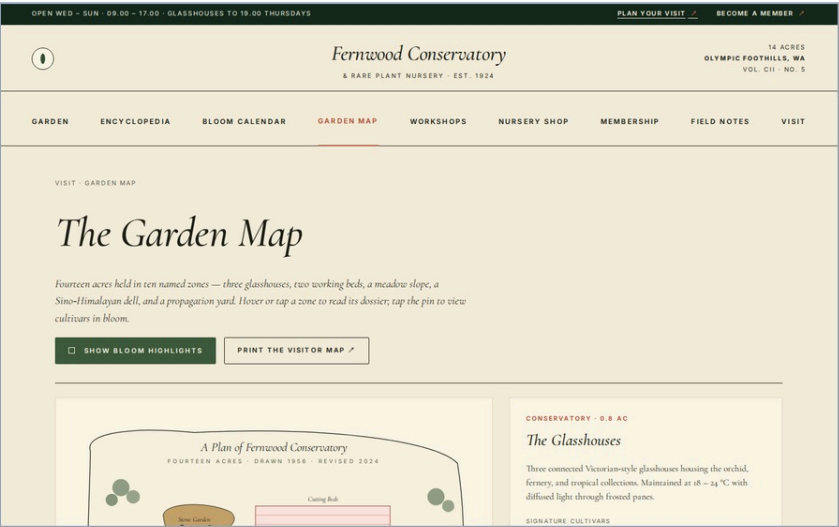
Map/highlight state shows the benchmark is not only static homepages.

Reference / oracle



capture score 0.810 · VB 0.951 · VB size 0.992 · block PM 0.606 · CSSOM 0.863
The candidate captures the same visible block system and highlighted state.

Candidate



Botanical garden map state: labels, cards, map area, and highlighted state structure mostly align.

This is the intended use case for visual block scoring. Once visible blocks correspond, the metric checks whether the same pieces exist with similar area, text, placement, and style.

capture score 0.810

VB 0.951

VB size 0.992

block PM 0.606

CSSOM 0.863

8. Evidence: text can match while geometry is wrong

Booking workflow exposes state/layout mistakes

Same contact step: text exists, but previous-step content remains and geometry changes.

Reference / oracle

The reference UI for 'Step 4 · Contact & deposit' features a top navigation bar with links for Home, Menu, Library, Events, Book, Tavern Pass, and Reviews. A progress bar at the top shows four steps: 1. Pick room, 2. Date & party, 3. Add-ons, and 4. Contact & deposit (the current step). The main content area is divided into two columns. The left column contains a form with fields for 'Your name' (Riley Tabletop), 'Phone' ((555) 100-2000), 'Email' (riley@example.com), 'Occasion (optional)' (Friends night), and a text area for 'Anything we should know?'. Below the form is a checkbox for 'I agree to the deposit terms...' and two buttons: 'Back' and 'Pay deposit & reserve'. The right column displays 'Your reservation' details: Room (not selected), When (2026-05-23 3h), Party (8 players), Estimated total (\$0), and Deposit due (\$0). A note at the bottom states 'Deposit applies to your final tab. Free reservation 48h ahead.'

capture score 0.534 · HTML 0.613 · VB text 0.939 · VB size 0.061 · BBox 0.026 · CSSOM 0.047
The task includes workflow states, not only polished landing pages.

Candidate

The candidate UI for 'Step 3 · Add-ons (optional)' features a top navigation bar with links for Home, Menu, Library, Events, Book, Tavern Pass, and Reviews. A progress bar at the top shows four steps: 1. Pick room, 2. Date & party, 3. Add-ons (the current step), and 4. Contact & deposit. The main content area is divided into two columns. The left column contains a form with fields for 'Your name' (Riley Tabletop), 'Phone' ((555) 100-2000), 'Email' (riley@example.com), and 'Occasion (optional)' (Friends night). The right column displays 'Your reservation' details: Room (not selected), When (2026-05-23 3h), Party (8 players), Estimated total (\$0), and Deposit due (\$0). Below the form is a checkbox for 'I agree to the deposit terms...' and two buttons: 'Back' and 'Pay deposit & reserve'. The right column displays 'Your reservation' details: Room (not selected), When (2026-05-23 3h), Party (8 players), Estimated total (\$0), and Deposit due (\$0). A note at the bottom states 'Deposit applies to your final tab. Free reservation 48h ahead.'

Tabletop booking contact state: the contact-form text is present, but the candidate keeps the previous add-ons step above it and changes the reservation-panel geometry.

This case explains why visual block cannot be reduced to text. The candidate includes much of the right form content, but the block area, local pixels, BBox, and CSSOM show that the state layout is wrong.

capture score 0.534

HTML 0.613

VB text 0.939

VB size 0.061

block PM 0.027

BBox 0.026

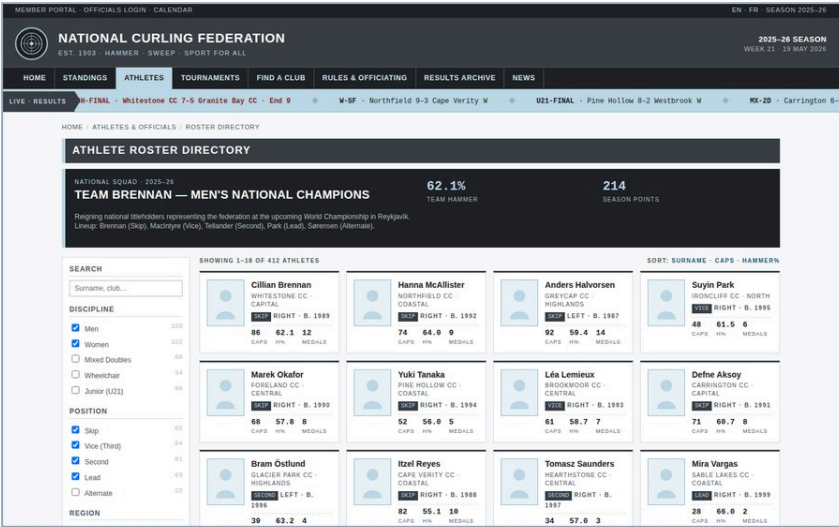
CSSOM 0.047

9. Evidence: local density matters

Sports federation: same domain, wrong density

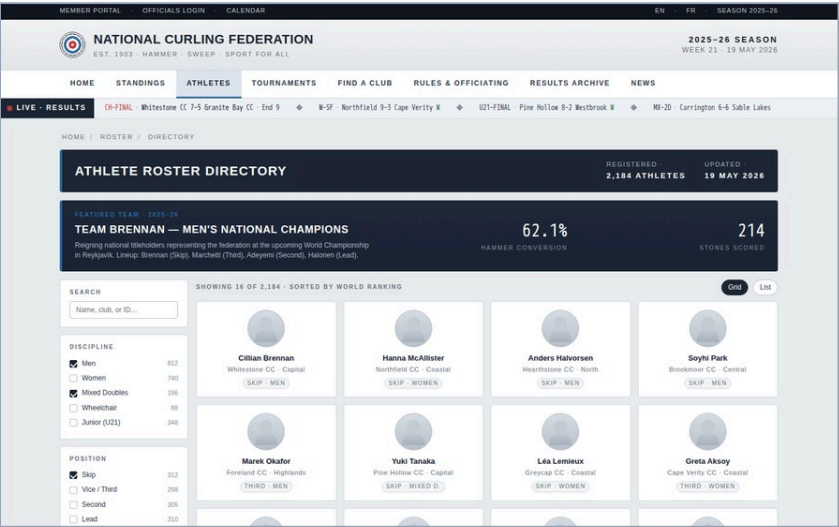
Athlete directory reveals card density, table hierarchy, and nav/header geometry drift.

Reference / oracle



trial reward 0.414 · athletes capture 0.218 · HTML 0.346 · BBox 0.119 · CSSOM 0.369
The agent understands the domain but redesigns the local system.

Candidate



Curling athlete directory: the page identity is correct, but card density, header geometry, filter/sidebar treatment, player count layout, and footer/page geometry differ.

This is the case for local fidelity metrics. Broad screenshot identity and visual-block aggregate can still recognize the page, but HTML, CSSOM, BBox, and blockwise pixelmatch correctly penalize density and layout drift.

reward 0.414

athletes capture 0.218

DreamSim 0.766

HTML 0.346

BBox 0.119

CSSOM 0.369

10. Evidence: agents reproduce vibe better than states

Booking workflow exposes state/layout mistakes

Same contact step: text exists, but previous-step content remains and geometry changes.

Reference / oracle

The reference booking workflow for 'Meeple & Mug' shows a four-step process: 1. Pick room, 2. Date & party, 3. Add-ons, and 4. Contact & deposit. The current step is 'Step 4 · Contact & deposit'. It features a form for user details (name, phone, email, occasion) and a summary box titled 'Your reservation' showing room, date, party size, and estimated total. A 'Pay deposit & reserve' button is at the bottom.

capture score 0.534 · HTML 0.613 · VB text 0.939 · VB size 0.061 · BBox 0.026 · CSSOM 0.047
The task includes workflow states, not only polished landing pages.

Candidate

The candidate booking workflow for 'Meeple & Mug' shows a three-step process: 3. Add-ons (optional) and 4. Contact & deposit. The current step is 'Step 3 · Add-ons (optional)'. It features three toggleable add-on cards: 'Snack platter', 'Dedicated host', and 'Themed setup'. A 'Next: contact' button is at the bottom.

Tabletop cafe: cheerful domain style is present, but multiple requested state captures are missing.

The agent is stronger at static page construction than interaction-state reproduction. Missing filters, toggles, accordions, selected nodes, and alternate views should remain first-class reward failures.

reward 0.556

coverage 0.850

missing 4/17 states

visual style plausible

11. Metric usefulness

Metric	Usefulness	Evidence	How to use it
Rendered HTML	Very strong	Trial corr 0.933	Best discriminator for content, section, and repeated DOM structure drift.
CSSOM	Very strong	Trial corr 0.840	Catches browser-computed style: typography, spacing, color, layout.
Visual block aggregate	Strong	Capture corr 0.587	Useful visible-block fidelity signal, but must be read with submetrics.
Visual block size	Strong	Capture corr 0.558	Tells whether the same visual block area exists. This catches missing/reshaped components.
Visual block text	Strong but incomplete	Capture corr 0.516	Good content-block signal; can overcredit when text appears in the wrong geometry.
Blockwise pixelmatch	Useful local check	Capture corr 0.407	Compares pixels inside matched blocks, avoiding many whole-page false positives.
BBox geometry	Useful but harsh	Capture corr 0.456	Good for card density and placement; not reliable alone.
VLM judge	Useful sanity signal	Trial corr 0.409	Sees human-facing quality, but can overrate plausible wrong structure.
DreamSim	Broad only	Compressed 0.747-0.862	Good catastrophic mismatch guard, weak final reward driver.
Global pixelmatch	Debug only	Trial corr ~0	Too sensitive to whitespace/page size and too forgiving when backgrounds dominate.
Masked CLIP	Do not use yet	Skipped/null	No evidence it adds useful signal; avoid CLIP forward passes for now.

Important

DreamSim and global pixelmatch are not wrong metrics; they are wrong as dominant reward drivers. They are useful as guards or diagnostics, while rendered HTML/CSSOM/visual block carry the real fidelity signal.

12. Visual block breakdown

Visual block is useful because it decomposes visible text/content blocks. The aggregate score is less informative than the submetrics.

Submetric	Range	Mean	Capture-score corr	Interpretation
Size	0.038-0.997	0.668	0.558	Whether the same visible block area exists.
Text	0.697-1.000	0.908	0.516	Whether matched blocks contain similar text.
Position	0.254-0.966	0.861	0.313	Whether the block lands in a similar location.
Text color	0.731-0.968	0.845	0.158	Mostly diagnostic; weak reward driver.
Block pixelmatch	0.006-0.713	0.316	0.407	Local pixel fidelity inside matched blocks.

What worked

Size and text were the strongest visual-block submetrics. Size catches whether the same visible block area exists; text catches whether the block content is similar.

What is niche

Text color is useful for diagnosis but weak as a reward driver. Blockwise pixelmatch is more useful than global pixelmatch because it focuses on matched local components.

13. Agent insights and reward implications

What the agent does well

- Creates plausible page scaffolds and routes.
- Captures broad art direction: palette, mood, hero/footer, domain feel.
- Builds recognizable components such as cards, grids, sidebars, dashboards, menus, and editorial sections.

Where it struggles

- Exact card density, spacing, row/column geometry, and local alignment.
- Rendered text/content structure and repeated DOM patterns.
- Stateful captures: filters, accordions, selected nodes, tabs, toggles, compare states.
- Custom diagrams/assets such as maps, brackets, rosters, botanical drawings, and audio panels.

Reward changes

- Keep rendered HTML and CSSOM central.
- Keep visual block, but always report size/text/position/color/block-pixelmatch.
- Use blockwise pixelmatch, not global pixelmatch, as the local pixel signal.
- Keep VLM as a sanity signal, paired with deterministic browser-rendered artifacts.
- Keep DreamSim small: good guard, weak fine-grained reward.
- Treat state misses as first-class reward events, not evaluator noise.

Final read

The reward is directionally working because it does not simply reward broad visual similarity. Weak candidates can look plausible, but final reward drops when rendered HTML, CSSOM, visual block size, bbox geometry, or state coverage fail.